Research Notebook — Emergent Geometry and Decoherence: Reflections on the Informational Origin of Spacetime

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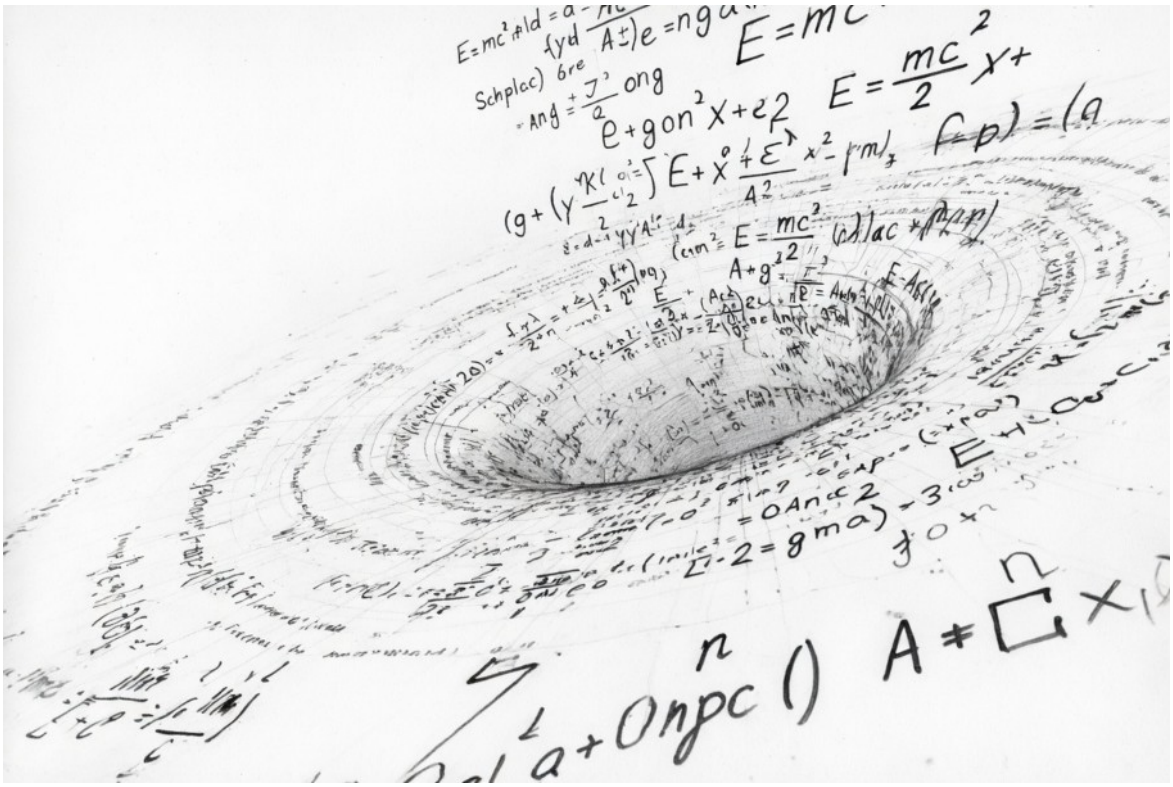
Working paper — Version 1.0 — Comments welcome



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Space and time do not exist. Not fundamentally. What we call geometry, distance, duration — are emergent properties, the result of the interaction between particles and the information structure they share. This document explores this hypothesis: space-time as a consequence, not as a backdrop.

🌟 SECTION 1 — Introduction and Motivation

It all began with a practical, somewhat silly question: can we use gravity as a source of free energy? The answer seems to be no (but I didn't give up!) — gravity takes back exactly what it gives. But the question led me to another, deeper one: what exactly is gravity? Following this thread — Rovelli (🫶), Einstein, Alice and her atomic clock (I'll introduce her later: section 2.4) — I arrived at a dizzying conclusion: space-time cannot be fundamental. It must emerge from something. So I turned the problem on its head. What if space didn't exist before particles? What if time wasn't a backdrop, but a result?

I should point out that, beyond the conceptual and heuristic intuition which is entirely my own, the mathematical and computational formalisation was carried out with the help of Copilot. Critical review, challenging questions and clarifications were provided by Deepseek, Gemini and Claude. I could not, on my own, have translated these ideas into a framework in the tradition of Van Raamsdonk, Verlinde or Zurek — fields far removed from my initial training. **Sections 2 to 5** provide a mathematical formalisation of the intuition outlined in the introduction. Part of this formalism was constructed with the help of AI tools — I do not claim to master every technical detail, but I have verified that each concept accurately reflects what I am trying to convey. Readers without a mathematical background may skip directly to **Section 6** without losing the thread.

🌟 SECTION 2 — Conceptual Framework

Let us consider a theoretical vacuum. Not the vacuum of a school classroom, not the absence of air — an absolute, theoretical vacuum. This vacuum would be both immensely small and immensely large, since... it is alone.

In this vacuum, a particle P1 appears. It ripples across its own N1 dimensions — these are basic postulates of quantum physics: a particle cannot 'not move', and particles are wave-like in nature. But here, neither space, nor time, nor gravity exist. Why? Because P1 is alone. And alone, nothing is defined.

(This is Carlo Rovelli's central idea in Helgoland: quantum mechanics is a theory of relationships, not of isolated objects. A particle has properties only in relation to something else. P1 alone in the void is like source code running without a screen.)

Now, let us assume that P2 appears, with its own N2 dimensions — not necessarily the same as P1's.

At that moment, the N1 dimensions of P1 and the N2 dimensions of P2 interfere — in the quantum sense of the term when they are close, in the sense of a projection when they are distant. From this interference emerges an intermediate space, a set of common intermediate dimensions — let's call them Px — on which P1 and P2 represent each other. This is not a pre-existing space in which they move. It is a space that arises from their encounter. The reference frames are not fixed by an external observer with a set square — they emerge from the interaction itself.

Now, let us imagine 10^80 particles appearing simultaneously — which, over an infinitely long time, is statistically inevitable. From the decoherence of all these interactions emerges an infinity of possible intermediate dimensions. Among them: a stable three-dimensional space.

Why precisely 3 dimensions? Paul Ehrenfest demonstrated this as early as 1917: only a 3-dimensional space allows for stable orbits, coherent waves, and atoms that do not collapse. Below or above this, physics diverges. 3D is not an arbitrary choice — it is a statistical attractor.

But our 3D universe is not alone. Coexisting around it are all the other possible dimensional averages: 1D, 2D, 4D, and ND universes. Some lack physical consistency. Some contain particles entirely perpendicular to our grid — invisible by design, just as 3D is to Pac-Man, no matter how he zooms.

With this intuitive framework established, the following sections formalise it mathematically. We define precisely what 'visible dimensions', 'private dimensions' and 'correlation' mean, and how we measure the emergence — and collapse — of a geometry.

This section sets out the fundamental ideas underpinning the central hypothesis: **that space-time geometry could emerge from the information structure between quantum systems.**

Here we present the key concepts, each accompanied by a physical justification, an intuitive diagram, and a minimal mathematical formulation.

2.1 Particles with visible and private degrees

Within the narrative framework outlined above, the 'proper dimensions' of each particle are divided here into two categories.

In this model, a particle is not an abstract point in a pre-existing space. It is a **composite system**, consisting of:

- **visible degrees**: those that contribute to the emerging geometry,

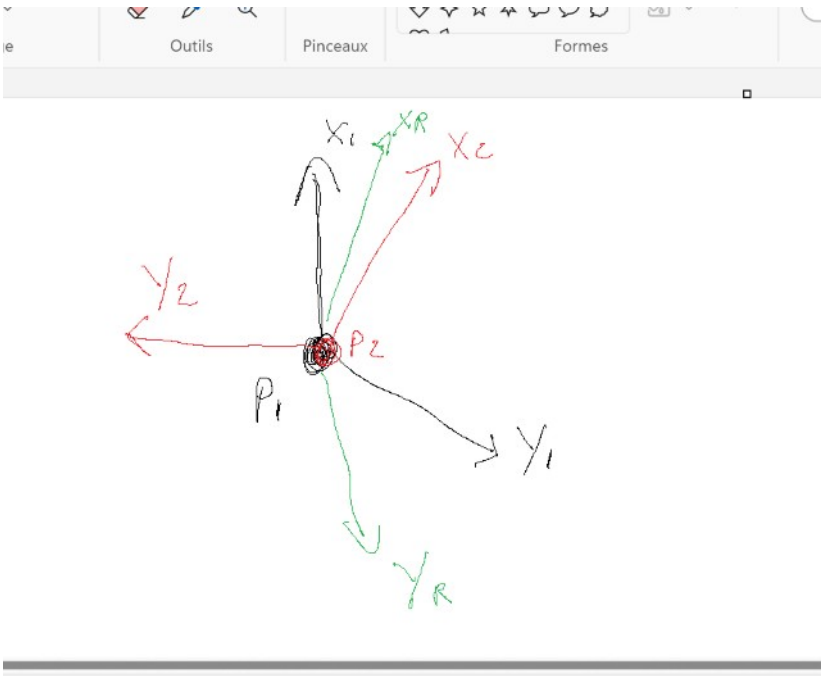
- **private degrees**: internal dimensions, not directly observable, but essential for storing and exchanging information.

Formally:

$$Hi = Hivis \otimes Hipriv$$

Private degrees play a role analogous to:

- **internal axes** possessed by each particle,
- **information reservoirs**,
- **hidden dimensions** that only become geometric when compared between particles.



This distinction is **central**: geometry is not within the particles, but in **the relationships between their visible parts**, modulated by their private parts.

2.2 Distance = correlation

In the story of P1 and P2, it was said that space arises from their encounter. But how do we measure this 'proximity' without pre-existing space? Through the information they share.

The key idea is simple:

Two systems are close if they share a great deal of information.

We replace the classical notion of distance with an operational measure:

$$distance(i, j) \propto f(Iij)$$

where Iij is the mutual information between the visible components of i and j.

```
High I_ij  →  many links  →  close particles
Low I_ij →  few links       →  distant particles
I_ij = 0    →  no correlation →  disconnection
```

This approach is consistent with:

- quantum physics (correlations = structure),
- quantum information (mutual information = dependence),
- holographic approaches (entanglement ↔ geometry).

2.3 Decoherence = geometric collapse

In our model, decoherence is not merely an enemy that destroys superpositions — it is what, on a large scale, has shaped our stable 3D universe. But locally, when it runs wild, it does the opposite: it collapses the geometry.

Decoherence is usually seen as noise that destroys superpositions. Here, it plays a **geometric** role:

- it erases fine correlations,
- it reduces the number of shared information directions,
- it flattens the emerging geometry.

```
Before decoherence:          After decoherence:

      (0)—(1)                      (0)     (1)
       |   |                        |       |
      (2)—(3)                      (2)     (3)

Rich geometry            Collapsed geometry
```

This is not merely a loss of information: it is an **effective reduction in dimension**.

2.4 Relational time

Time is not an absolute parameter. It emerges from the synchronisation of correlations between systems. It is time to introduce Alice.
Alice is a human locked in a room with an atomic clock (a cesium clock). Both count the 9,192,631,770 oscillations to deduce that a second has just elapsed and age normally. Now, Bob appears, on an Earth moving away from Alice at a fraction of the speed of light. Einstein taught us that they will age differently relative to one another — but for each of them, the cesium atom is still doing its job. 9 billion oscillations. Spot on.
What changes is the fabric itself — warped below the Planck scale by the mere fact that they exist in relation to one another.
Two interacting particles define a mutual 'present'. A correlated network defines an arrow of time.

Formulation:

$$trel(i,j) = fonctiondeladynamiquedescorrélationsIij(t)$$

ASCII intuition:

Code

No correlation → no shared time

Stable correlation → shared time

Dynamic correlation → arrow of time

2.5 Why this framework is physically consistent

This model is part of a line of modern thinking (So, I didn't start from this work — I arrived at it later. Upon discovering it, I had confirmation that others had already looked in that direction...):

- **Zurek**: decoherence selects stable states (e-selection).
- **Van Raamsdonk**: entanglement constructs the connectivity of space.
- **Verlinde**: gravity can emerge from information.
- **Penrose / Diósi**: gravity and state reduction are linked.

But it differs from these in one crucial respect:

Here, geometry is not derived from a holographic theory or a continuous field, but from a local network of correlations between finite systems.

It is a **bottom-up**, operational approach, compatible with numerical simulations.

🌟 SECTION 3 — Mathematical formalism

This section presents the mathematical tools that enable the conceptual framework to be transformed into an operational formalism. The aim is not to be exhaustive or overly technical, but to provide **robust definitions**, **comprehensible equations**, and **physical justifications** for each choice.

3.1 Quantum states and reduction

We consider a set of particles, each composed of two subspaces:

$$Hi = Hivis \otimes Hipriv$$

The overall state of the system is a density matrix:

$$\rho(t)$$

But the observer has access only to the observable degrees of freedom. We therefore define the **reduced** state:

$$\rho vis(t) = Trpriv\rho(t)$$

This partial trace operation:

- eliminates the hidden degrees of freedom,
- introduces entropy (mixing),
- and paves the way for the concept of visible correlation.

Interpretation: Our universe is not a filter that masks hidden dimensions — it is a stable projection. Only the dimensions capable of expressing themselves within it construct our reality. The others are not lost; they are simply perpendicular to our framework, and therefore silent to us. Private dimensions cannot manifest themselves in our three-dimensional system. This filter creates missing information, and thus entropy.

3.2 von Neumann entropy

The quantum entropy of a state ρ is:

$$S(\rho) = -Tr(\rho log 2 \rho)$$

It measures:

- the degree of mixing,
- uncertainty,
- the amount of missing information.

Special cases:

- pure state → S=0
- maximized state → S=log2d (where d is the dimension)

3.3 Mutual information: the geometric compass

This is where the intuition behind P1 and P2 becomes measurable: mutual information Iij is precisely what quantifies the 'encounter' between two particles.

For two visible particles i and j, we define:

$$Iij = S(\rho i) + S(\rho j) - S(\rho ij)$$

This quantity measures **all** the shared information:

- classical correlations,
- quantum entanglement,
- structural dependencies.

Why is this important?

Because in this model:

<u>Geometry is the map of correlations.</u>

So Iij is our **inverse distance**.

ASCII:

Code

I_ij large → nearby particles

I_ij small → particles far apart

I_ij = 0 → no relationship → disconnection

3.4 Overlap matrix W: directional normalization

The Iij matrix gives the raw correlations. But to understand the **relative** structure, we normalize by row:

$Wij = Iij \sum kIik$

Why this normalization?

- It makes the rows comparable;
- it expresses *the proportion* of information in i that comes from j,
- it eliminates scale effects,
- it highlights the dominant directions.

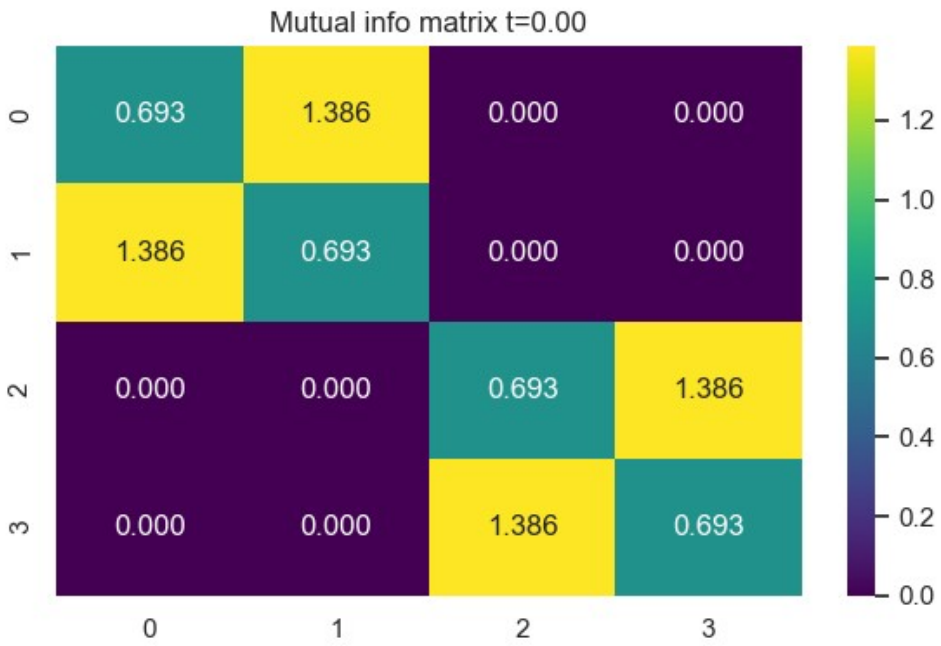


Figure 1: t=0: initial geometry

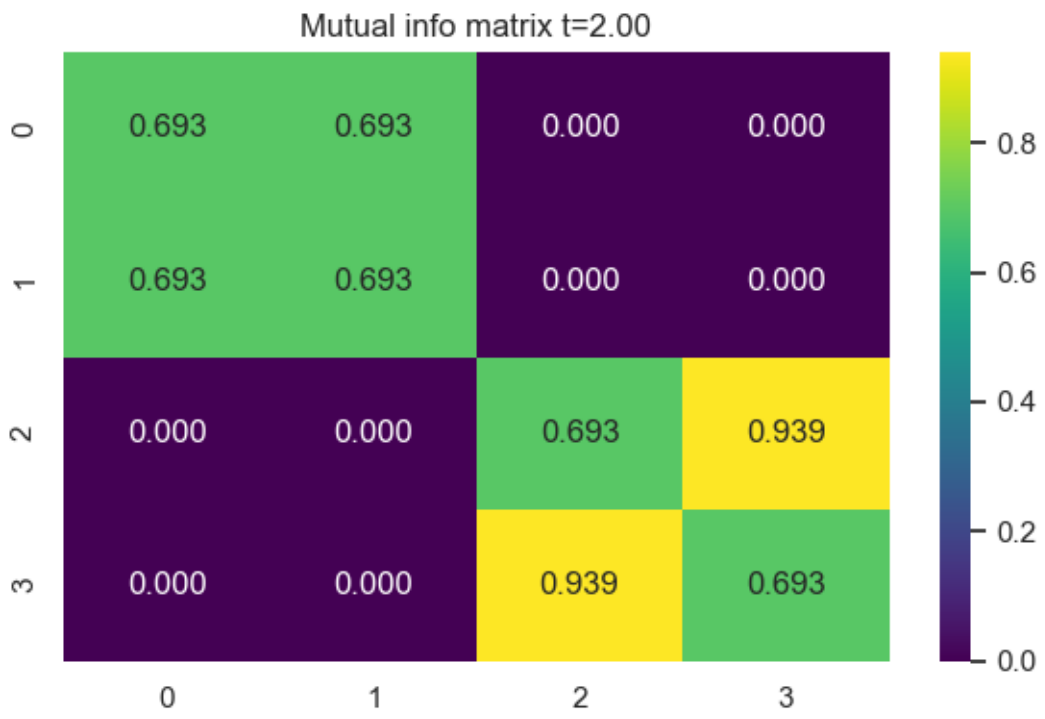


Figure 2: t=2: after decoherence"

Interpretation:

$$W_{ij} = partofinformationwhichdependofj$$

If a line is flat → the particle no longer has a directional structure → geometric collapse.

3.5 Effective dimension D_eff: measuring geometric richness

Having correlations is good. But how many independent directions do these correlations create? This is what D_eff measures — the dimensional richness of the emerging geometry

We diagonalise W:

$$Wvk = \lambda kvk$$

The eigenvalues λk indicate **how many independent information directions** exist.

We define:

$$Deff = \left(\sum k\lambda k\right)2 \sum k\lambda k\,2$$

This is the **participation ratio**, a standard measure in statistical physics.

Interpretation:

if all λk are equal → maximum Deff

- if a single one dominates → Deff ≈ 1
- if decoherence erases the directions → Deff drops

```
Before decoherence: λ = [0.25, 0.25, 0.25, 0.25] → D_eff = 4
After decoherence: λ = [1, 0, 0, 0]              → D_eff = 1
```

This is a **quantitative measure of geometric collapse**.

3.6 Informational distance: from information to geometry

We now have a measure of correlation. We need to transform it into a distance — because that is the language geometry understands. The more correlated two particles are, the closer they are. The exponential function performs this translation.

To convert Iij into a distance, we use:

$$Dij = \exp\left(-\alpha Iij\right)$$

where $\alpha > 0$.

Why the exponential?

- it ensures $Dij \in$,
- it amplifies the differences between strong and weak correlations,
- it is stable for embeddings,
- it avoids division by zero.

ASCII:

Code

I_ij = 1.0 → D_ij ≈ 0.37

I_ij = 0.1 → D_ij ≈ 0.90

I_ij = 0 → D_ij = 1 (maximum distance)

The parameter α controls the sensitivity: a large α amplifies the differences, a small α smooths them out. Its optimal value remains an open question.

3.7 Geometric embedding: reconstructing the space

All these distances have been calculated — but where are the particles in space? MDS embedding answers this: it places the particles in a 2D space whilst respecting the informational distances as closely as possible. This is not THE geometry — it is ONE compatible geometry.

Using the distance matrix Dij, we apply an embedding algorithm (MDS) to obtain coordinates:

$$Dij \quad xi \in R2$$

The aim is not to find *the* geometry, but **a geometry compatible** with the informational distances.

We then calculate the **stress**:

$$\sigma = \sum i<j ¿¿$$

- low stress → the emerging geometry is coherent
- high stress → the structure is not metric (collapse, chaos, transition)

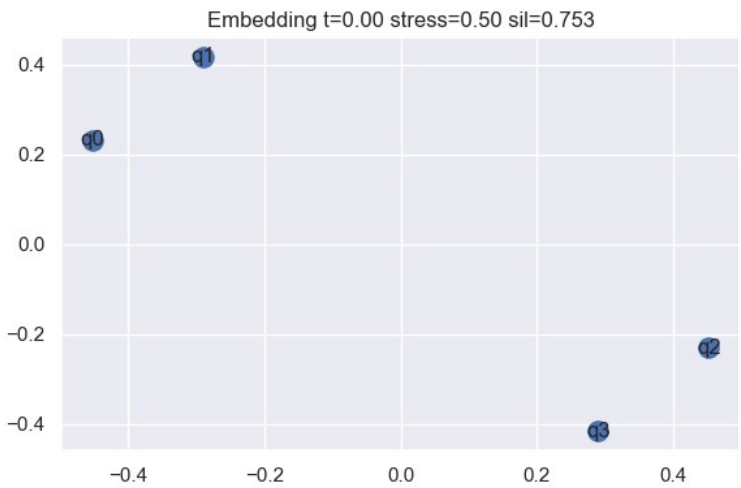


Figure 3: MDS embedding at t=0 — particles 0 and 1 are close (strongly correlated), 2 and 3 form a second isolated cluster. Stress=0.50 indicates a geometry in the process of forming.

3.8 Summary: how it all fits together

Here is the complete sequence:

Code

```
Global state ρ(t)
    ↓ partial trace
ρ_vis(t)
    ↓ entropies
S(ρ_i), S(ρ_ij)
    ↓ information
I_ij
    ↓ normalisation
W_ij
    ↓ spectrum
D_eff
    ↓ exponential
D_ij
    ↓ embedding
x_i(t), stress
```

Every step has a clear physical meaning. Every quantity is numerically measurable. And together, they allow us to **reconstruct an emergent geometry**.

This is the complete chain — from the raw quantum state to a visible geometry. Each step has a physical meaning; every quantity is measurable. This is not magic: it is translation.

🌟 SECTION 4 — Toy model

This section presents a toy **model** that is simple yet sufficiently rich to illustrate the central idea: **geometry can emerge from the information structure between quantum systems, and decoherence can destroy it.**

The aim is not to reproduce real physics in all its complexity, but to construct a system:

- controllable,
- numerically simulable,

- interpretable,
- and capable of demonstrating the key phenomena: **correlations → geometry**, **decoherence → geometric collapse.**

4.1 Structure of the system

Ancillas is the technical term given to the private degrees in this model — the invisible dimensions of each particle, those that do not participate directly in the emergent geometry but modulate it from within. In the diagram above, vis0 is what we see of particle 0; a0 and b0 are what it keeps to itself.

We consider a set of **four visible particles**, each accompanied by **two or three private ancillas**.

Why 4 particles? This is the minimum required to construct two correlated pairs (0↔1 and 2↔3) and observe how decoherence affects them independently. It is a minimalist laboratory — not a complete model of the universe. Similarly, the 2 ancillas per particle are a choice made for simplicity: in the general theory, each particle possesses N private dimensions. Here we take 2 to keep the simulation manageable.

Formally:

$$H= \quad i=03\left(Hivis \otimes Hipriv\right)$$

Each visible particle is a qubit. Each private ancilla is also a qubit.

Code

```
Particle 0: [vis0] — [a0] — [b0]

Particle 1: [vis1] — [a1] — [b1]

Particle 2: [vis2] — [a2] — [b2]

Particle 3: [vis3] — [a3] — [b3]
```

The ancillas serve as **reservoirs of internal information**, modulating the structure of the visible correlations.

4.2 Initial state: correlated structure

Before running the simulation, a starting state must be chosen. A state that is too simple would show nothing of interest — a state that is too complex would be unmanageable. We seek the happy medium: enough correlations for decoherence to have something to destroy

The initial state must:

- contain visible correlations,
- contain hidden correlations,
- be sufficiently rich for decoherence to distort it.

We use:

✔ Local GHZ on each block (visible + ancillas)

$$|GHZ\rangle i = |000\rangle + |111\rangle 2$$

This creates a **coherent internal structure** for each particle.

✔ Bell pairs between visibles

$$|\Phi+\rangle 01 = |00\rangle + |11\rangle 2$$
$$|\Phi+\rangle 23 = |00\rangle + |11\rangle 2$$

Code

(0)==(1) (2)==(3)

Bell Bell

GHZ and Bell are two standard ways of creating strongly entangled particles — that is, particles whose states are linked, even at a distance. This is the starting point: a system rich in correlations, which we will then allow to degrade under the effect of decoherence

✔ Slight random perturbation

To avoid perfect symmetries (which mask the effects of decoherence), we add very weak phase noise.

4.3 Dynamics: Lindblad master equation

The Lindblad equation is the formula that describes how our system degrades over time. The first term represents the intrinsic physics. The second term represents the decoherence that erodes the correlations — like rust on metal. The larger γ is, the faster the rust spreads.
In our toy model, we vary γ over several orders of magnitude to observe the three regimes: stable geometry, transition, collapse

The dynamics are given by:

$$d\rho dt = -i[H,\rho] + \sum k(Lk\rho Lk\dagger - 12\, Lk\dagger Lk,\rho)$$

In the toy model:

- we can take H=0 (purely dissipative dynamics),
- or add a simple coupling between the visible states.

We take H=0 for simplicity: we wish to observe pure decoherence, without any intrinsic dynamics complicating the interpretation.

✔ Phase decoherence (phase damping)

$$L_k = \gamma_k Z_k$$

where Zk is the applied Pauli Z operator:

- either to the quanta,
- or to the ancillas,
- or to both (the most interesting case).

Code

Decoherence on visible light:

vis0 — vis1 — vis2 — vis3

Decoherence on ancillas:

a0,b0; a1,b1; a2,b2; a3,b3

The key parameter is **γ**, the decoherence rate.

We explore several orders of magnitude:

γ∈[10−6,10+1]

4.4 Simulation pipeline

At each time step, the simulation runs the same sequence of calculations — the one constructed in Section 3, applied in real time:

✔ 1. Reduced states

ρi(t), ρij(t)

✔ 2. Entropies

S(ρi), S(ρij)

✔ 3. Mutual information

Iij(t)

✔ 4. Overlap matrix

Wij(t)

✔ 5. Effective dimension

Deff(t)

✔ 6. Information distances

$D_{ij}(t)=\exp(-\alpha I_{ij})$

✔ 7. Geometric embedding

$D_{ij} \rightarrow x_i(t)$

✔ 8. Stress

$\sigma(t)$

✔ 9. Half-life

$\langle I(t_{1/2})\rangle = 12 \langle I(0)\rangle$

This is the same sequence as in Section 3.8 — but this time it runs, step by step, whilst decoherence does its work.

4.5 What this model shows

Even though it is only a toy model, it allows us to observe:

✔ A regime of stable geometry

For very small γ, the correlations persist. The reconstructed geometry is stable.

✔ A transition regime

For intermediate values of γ, the correlations gradually break down. The geometry becomes distorted, then fragments.

✔ A geometric collapse regime

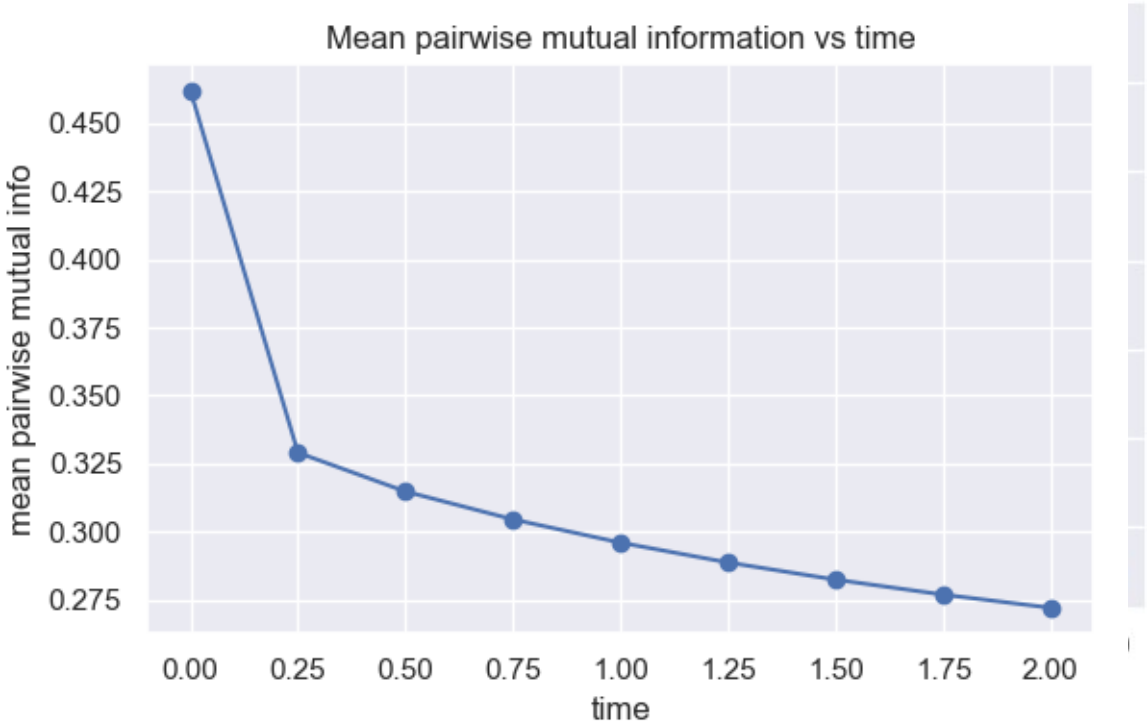
For high γ, correlations disappear rapidly. The geometry flattens, D_{eff} drops, and stress increases.

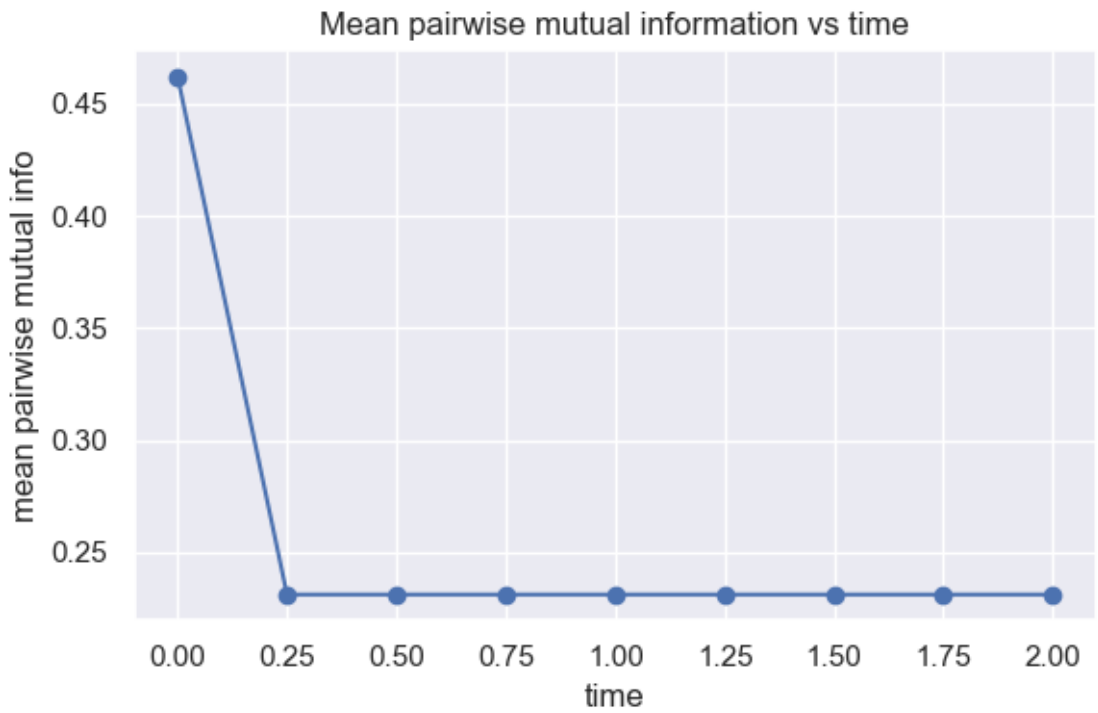
Code

```
low γ:   o—o—o—o   (coherent structure)

Medium γ:    o—o  o—o   (deformation)

High γ:     o  o  o  o   (collapse)
```


Mean pairwise mutual information vs time
mean pairwise mutual info
0.450
0.425
0.400
0.375
0.350
0.325
0.300
0.275
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
time


Mean pairwise mutual information vs time
mean pairwise mutual info
0.45
0.40
0.35
0.30
0.25
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
time

4.6 Why this model is relevant

A toy model, by definition, does not claim to replicate the universe. But this one does something useful: it shows that the idea works, even on a small scale.

Although simple, this model:

- captures the essence of the idea,
- is fully simulable,
- allows hypotheses to be tested,
- enables the visualisation of the emergence and collapse of the geometry,
- provides quantitative diagnostics (I_ij, D_eff, stress),
- paves the way for more realistic models (MPS, MERA, tensor networks).

It is a conceptual laboratory. Not the final theory — proof that the direction is viable.

🌟 SECTION 5 — Expected results

In this section, I simply look at what the simulations yield when the mutual information matrix Iij evolves under the effect of decoherence controlled by a parameter γ.
I am not trying to "prove" anything: I am an amateur exploring an idea, tinkering with a model, and observing what it reveals.
The aim is not to prove — it is to observe.

The aim is to see how the emerging geometry reacts depending on whether the decoherence is very weak, moderate or very strong.
We will therefore look at three representative cases.

5.1 Case 1: Very small γ — stable geometry

When γ is very small, decoherence has virtually no effect. The Iij matrix remains virtually identical from start to finish, and the mean of the correlations does not change.

- The initial heatmap shows two strongly correlated pairs.
- The final heatmap is… the same.
- The mean(I) curve is flat.

In other words: **the geometry does not deform.** It remains stable, as if nothing were happening.

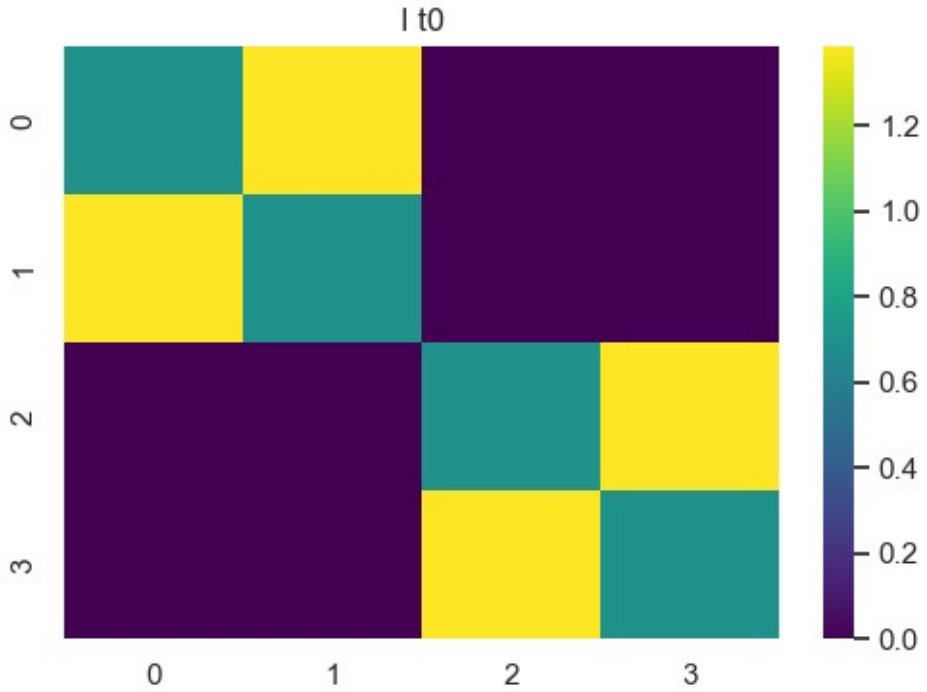


Figure 4: Initial heatmap ($\gamma = 6.329\times10^{-21}$). Initial correlation structure: two strongly linked pairs, no cross-correlations. The geometry is perfectly coherent.

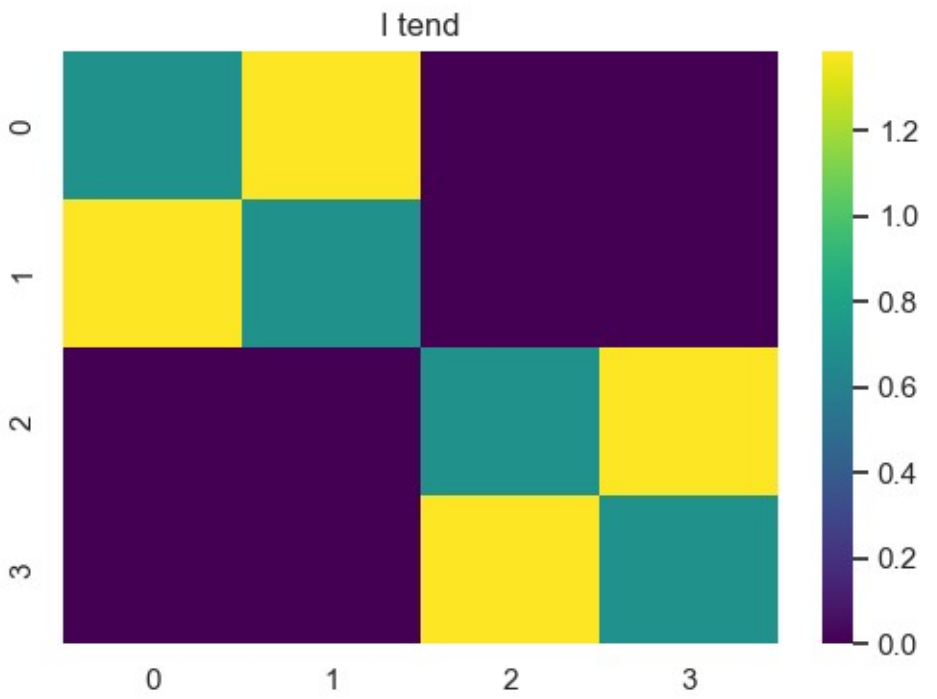


Figure 5: Final heatmap ($\gamma = 6.329\times10^{-21}$). No notable change: the final matrix is identical to the initial one. The decoherence is too weak to alter the geometry.

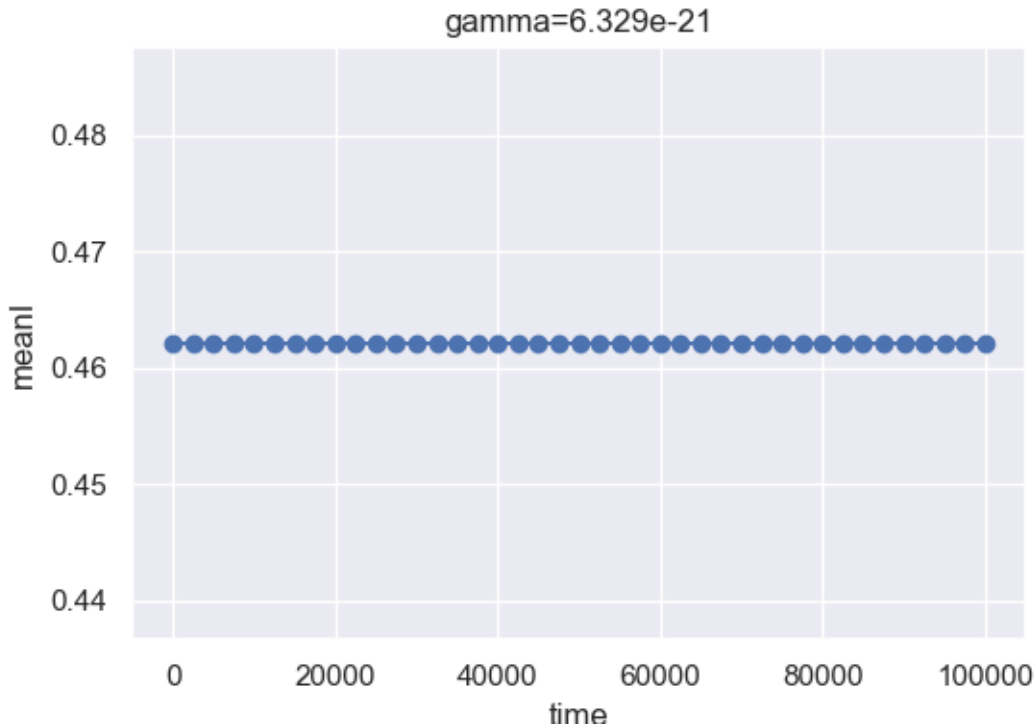


Figure 6: mean(I) over time ($\gamma = 6.329\times10^{-21}$). The mean of the correlations remains perfectly stable. The geometry does not deform.

5.2 *Case 2: intermediate γ — gradual transition*

Here, decoherence begins to take effect, but without disrupting everything all at once. We can clearly see that the correlations are decreasing, but slowly.

And this is where your vivid descriptions come into their own:

> ***Interpretation: decoherence acts as a force that 'pulls' on the geometry, distorting and fragmenting it.*** *We are not yet in a state of collapse, but we are no longer in a state of stability either.*

This behaviour corresponds to what we might call an ***informational phase transition****:*

> ***An informational phase transition is the moment when the system shifts from one state to another — like water changing from liquid to vapour, but here it is the geometry that changes state.***

The figures illustrate this clearly:

- *The initial heatmap resembles that of the stable case.*
- *The final heatmap shows weakened clusters, but they are still recognisable.*
- *The mean(I) curve descends gently.*

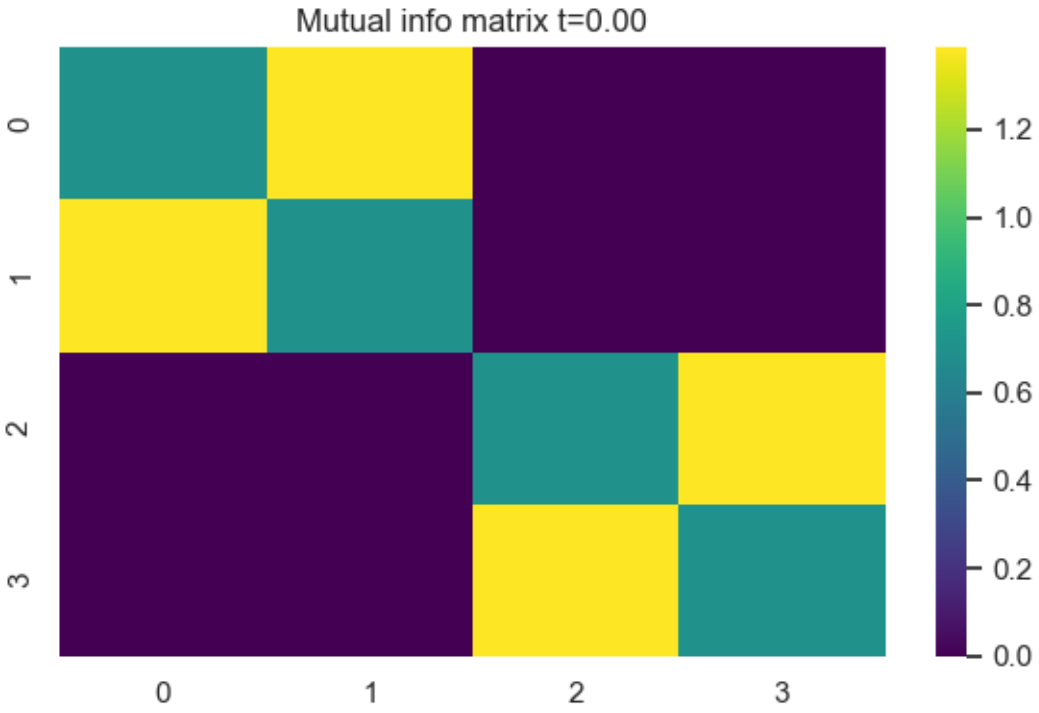


Figure 7: Initial heatmap ($\gamma = 2.359\times10^{-2}$). Initial structure similar to the stable case: two well-defined correlated blocks.

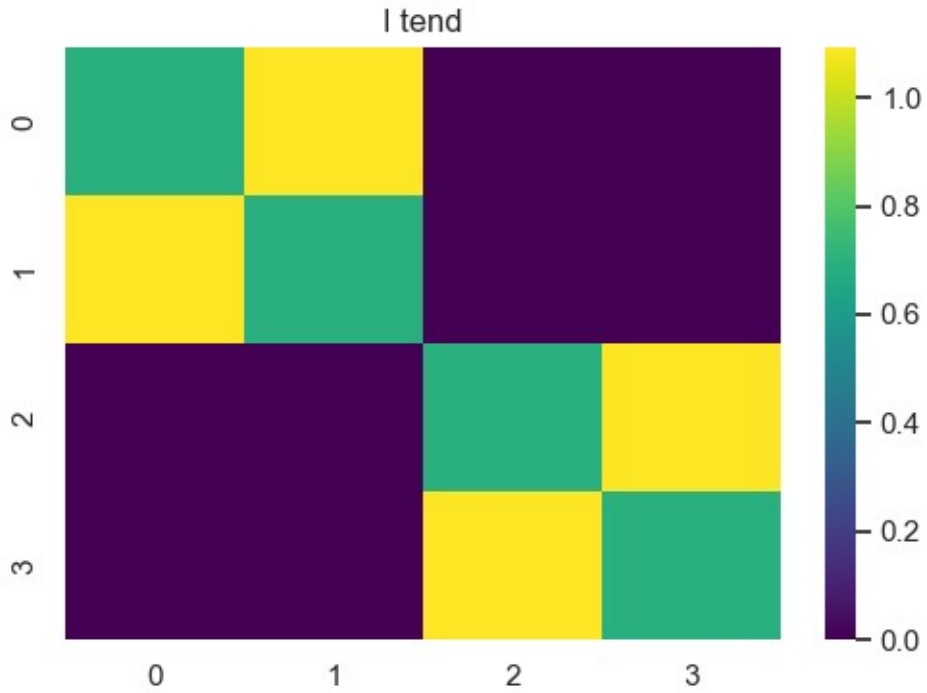


Figure 8: Final heatmap ($\gamma = 2.359\times10^{-2}$). The clusters are weakened: decoherence 'pulls' at the geometry, distorting and fragmenting it.

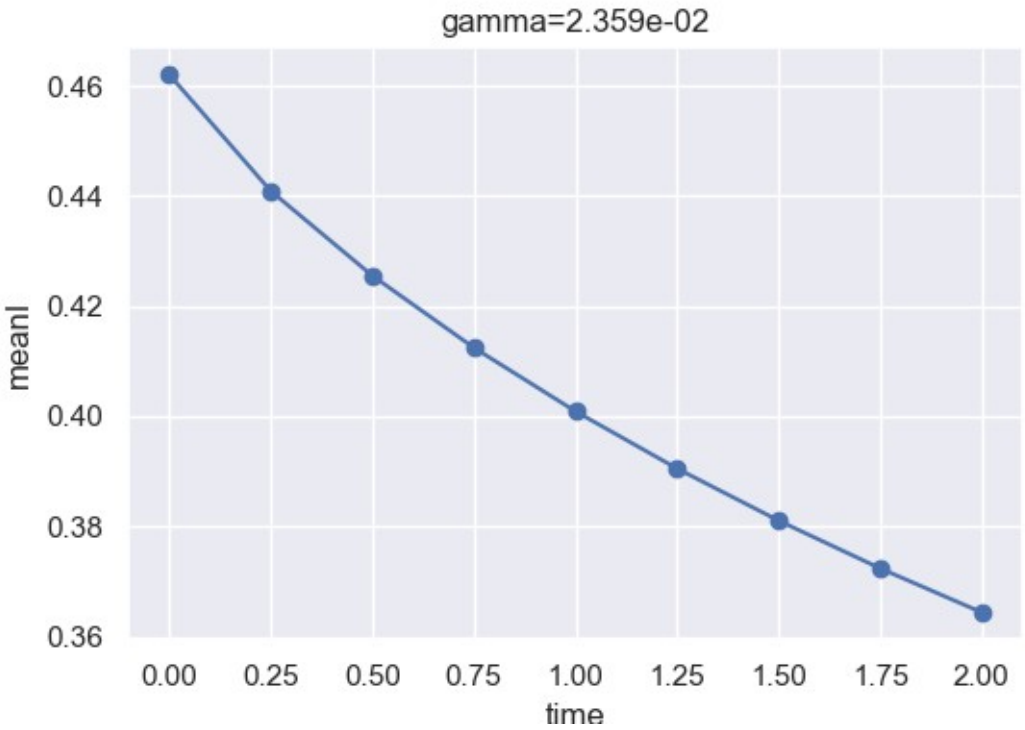


Figure 9: mean(I) over time ($\gamma = 2.359\times10^{-2}$). Slow and steady decline: the onset of an informational phase transition.

5.3 Case 3: very high γ — abrupt collapse

When γ becomes very large, decoherence overwhelms everything in an instant.

- The initial heatmap is normal.
- The final heatmap is almost empty.
- The mean(I) curve drops suddenly, then remains at rock bottom.

Here, there is no longer a gradual transition: **the geometry loses all coherence in an instant.**

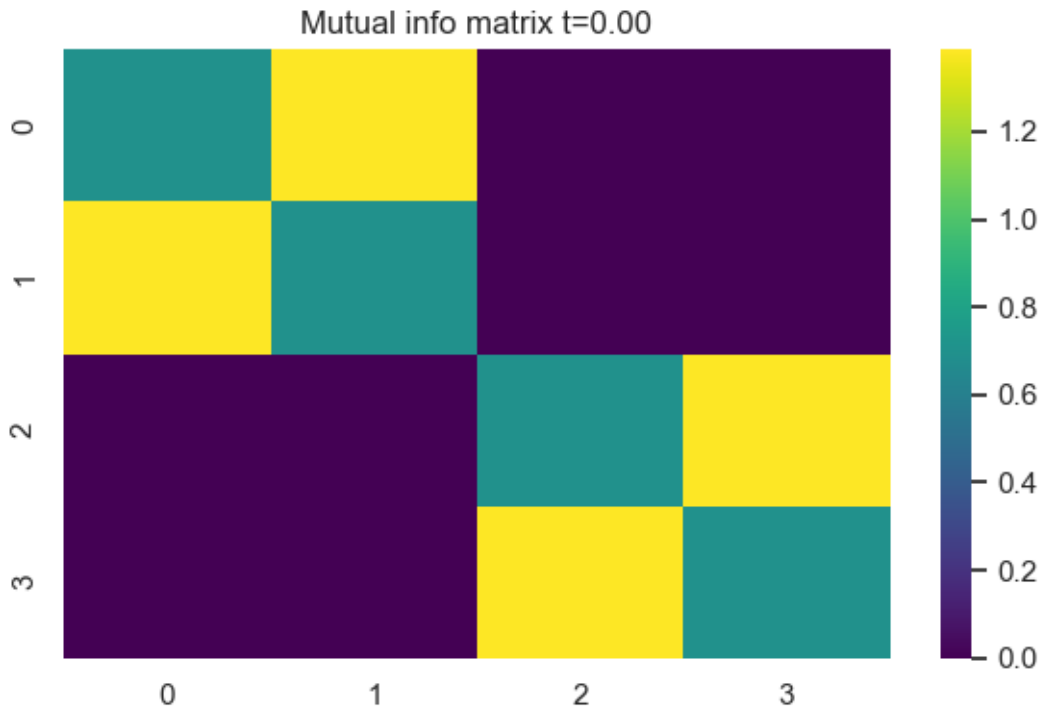


Figure 10: Initial heatmap (γ = 3.864×10³). Normal initial structure, comparable to previous cases.

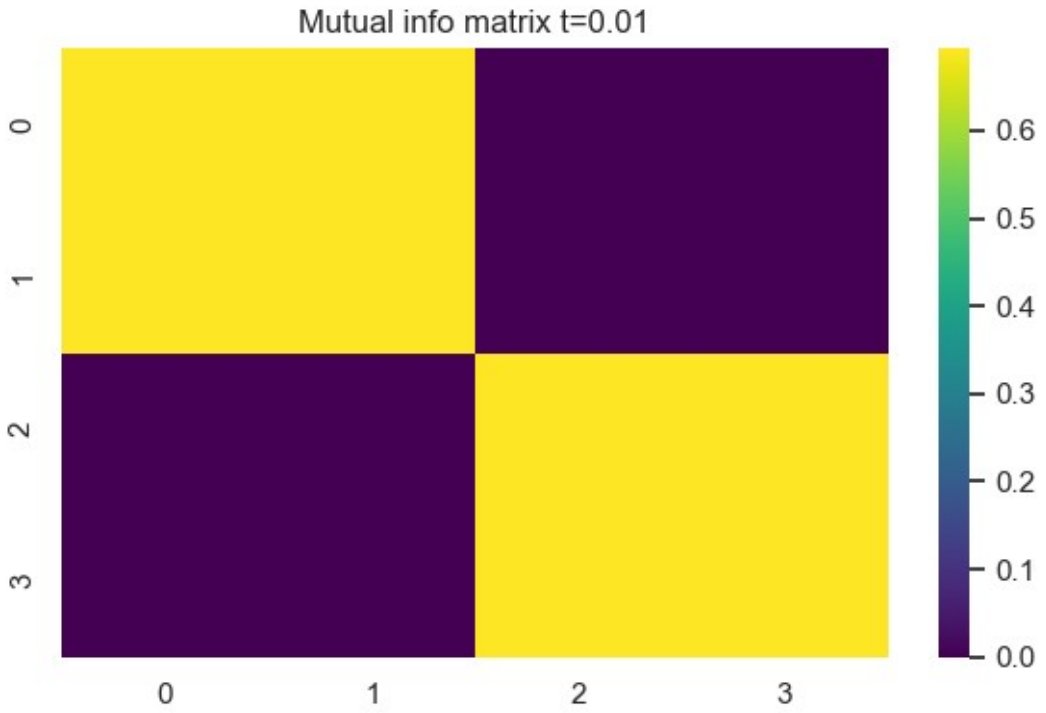


Figure 11: Final heatmap (γ = 3.864×10³). Total collapse: correlations disappear almost entirely. The geometry loses all coherence.

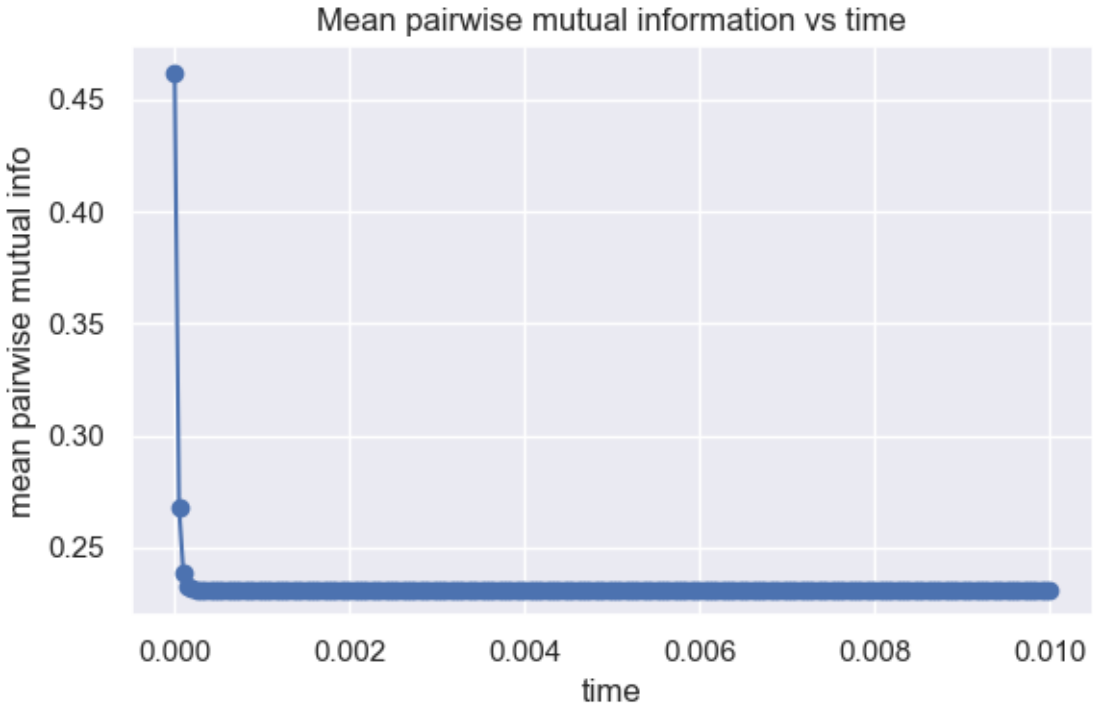


Figure 12: mean(I) over time ($\gamma = 3.864\times10^3$). Instantaneous drop in the mean of the correlations: abrupt collapse.

5.4 Summary: $t_{1/2}$ as a function of γ

To summarise all this, I have plotted the half-life of the correlations as a function of γ. The three regimes are clearly visible:

- very low γ → $t_{1/2}$ infinite (no decay)
- Intermediate γ → $t_{1/2}$ begins to decrease
- very high γ → $t_{1/2} \approx 0$ (immediate collapse)

We can therefore see a genuine **informational phase transition** emerging: when γ is low, the geometry remains coherent; when γ becomes too high, it collapses almost instantly.

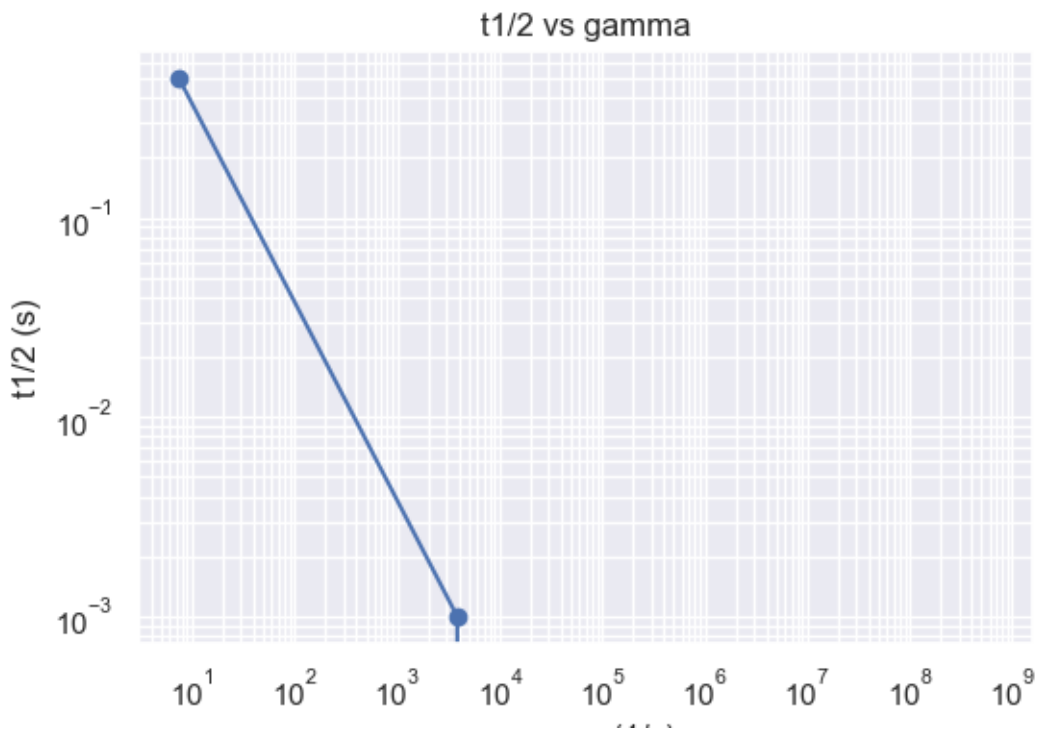


Figure 13: Half-life of correlations as a function of γ. The three regimes are clearly evident: stability ($t_1/_2$ infinite), gradual transition, followed by immediate collapse for high values of γ.

🌟 SECTION 6 — Cosmological Interpretations

We are now entering my territory. The previous sections have established the formalism — here, we apply it to the big questions. Not to replace Einstein or standard cosmology, but to propose a different interpretation: what if certain mysteries of the universe were informational effects rather than fundamental properties of space-time?

> **Spacetime geometry could emerge from the information structure between quantum systems.**

If this hypothesis is correct, then certain cosmological phenomena could be interpreted as **informational effects**, rather than as fundamental properties of space-time.

6.1 The Big Bang as an informational transition

In the standard model, the Big Bang is a singularity — a mathematical wall where the equations break down. Within this framework, we can propose something more satisfactory: the Big Bang is not a beginning in time, it is the beginning of time itself — the moment when the first correlations appear and give rise to space.

Within our framework, we can propose a different interpretation:

✔ Before the Big Bang

Particles (or proto-systems) exist, but **share no visible correlations**. Private degrees dominate. There is no emerging geometry.

Code

```
Before:

[•]  [•]  [•]  [•]

No correlation → no space
```

These proto-systems exist in the quantum sense of the term — that is, potentially everywhere at once, in a superposition of all possible states. As long as no interaction correlates them, they are 'nowhere' in any definite sense. Space does not yet exist to localise them.

✔ Transition

A mechanism (interaction, correlated decoherence, internal coupling) suddenly creates visible correlations.

Iij(t) increases abruptly

Geometry emerges **all at once**, like a percolation.

Code

```
Transition:

[•]—[•]  [•]—[•]

Correlations → beginning of space
```

✔ After

Correlations stabilise, geometry becomes coherent, the Universe "expands" because **the information structure becomes more complex**.

Code

```
After:

[•]—[•]

|    |

[•]—[•]

Stable geometry
```

Interpretation:

<u>The Big Bang is not an explosion in space. It is the emergence of space through the emergence of correlations.</u>

6.2 Dark matter: invisible private information

50 years of research, detectors buried kilometres underground, giant colliders — and still nothing. Not a single particle of dark matter has been detected. What if it is because we are looking in the wrong place — in visible space — when the answer lies in hidden information?

In this context, we can propose an alternative:

> **Dark matter would be the geometric effect of private correlations inaccessible to visible observers.**

Private degrees:

- store information,
- interact with the visible,
- alter the overall information structure,
- but are not directly measurable.

Mathematically:

$$I_{ij}^{total} = I_{ij}^{vis} + I_{ij}^{priv}$$

The observer sees only I_{ij}^{vis}, but the geometry depends on **everything** I_{ij}^{total}.

ASCII:

Code

```
Visible:

(0)——(1)


Private (hidden):

 a0——a1


Total effect:

(0)——(1)  ← appears closer than expected
```

Interpretation: Dark matter could be a **<u>geometry induced by hidden correlations</u>**.

This is exactly what the private degrees — the ancillas — do in the model: they influence the geometry without being directly observable. Dark matter would be the geometric shadow of dimensions that we cannot see, because they are perpendicular to our frame of reference.

6.3 Dark energy: informational pressure

Dark energy is the mystery of mysteries — a force that accelerates the expansion of the universe and which nobody knows what it is. In this framework, the answer is almost brutally simple: **the universe is expanding because it is forgetting.**

Within this framework, we can propose an elegant interpretation:

> **Dark energy would be the natural tendency of quantum systems to lose their visible correlations under the effect of decoherence.**

As visible correlations diminish:

- informational distances increase,
- the geometry 'stretches',
- space appears to expand more rapidly.

Mathematically:

$$Dij(t) = \exp\left(-\alpha Iij(t)\right)$$

If Iij(t) decreases slowly but overall:

Dij(t) increases everywhere

Code

Before:

o—o—o—o

After slow decoherence:

o o o o

Apparent expansion

Interpretation: Dark energy is thought to be an **informational expansion**, not a mysterious force.

6.4 Emergent gravity: curvature = information gradients

Einstein said: mass curves space. But why? In this framework, the answer is: because mass is a concentration of information. And since space is made of information, it is mathematically forced to contract where the information is dense. Gravity is not a force — it is a gradient:

Gravity is an information gradient.

If one region has more internal correlations than another:

- the informational distances contract,
- trajectories curve,
- particles 'fall' towards areas of high correlation.

Highly correlated zone:

```
 •••
•   •
 •••
```

→ attracts other systems

This is consistent with:

- Verlinde (entropic gravity),
- Van Raamsdonk (entanglement ↔ connectivity),
- Jacobson (Einstein = thermodynamics).
- These three physicists arrived at similar conclusions via different paths. This model follows in their footsteps — but from the bottom up, starting from local correlations rather than global thermodynamics.

It is the same mechanism as P1 and P2 — but on a macroscopic scale. A star consists of 10^57 particles whose internal correlations create an information gradient so dense that anything passing nearby 'falls' towards it.

6.5 Arrow of time: visible loss of information

Why does time flow in one direction and not the other? This is one of the deepest questions in physics. In this context, the answer is straightforward: because correlations only diminish. We cannot 'turn back' time because we cannot recreate correlations that have been erased by decoherence. Time is irreversible because forgetting is irreversible.

The arrow of time could be interpreted as:

$ddtIijvis < 0$

Decoherence:

- erases visible correlations,
- increases entropy,
- creates temporal asymmetry.

Code

```
t0: o—o—o—o

t1: o—o  o—o

t2: o  o  o  o
```

<u>The arrow of time would therefore be an arrow of decoherence.</u>

This is the natural conclusion of the relational concept of time set out in 2.4: Alice and Bob do not age over time — they age with it, at the rate of the decoherence of their own correlations.

🌟 SECTION 7 — Open questions and personal intuitions

The central idea of this work is simple: distance may be nothing more than a measure of correlation. And if that were true, then space itself would be nothing more than a map of our quantum relations.

In this view, gravity does not exist as a force — it is present because the density of information warps the very fabric of space-time, making our representation of the universe fundamentally non-uniform.

These are not certainties. They are intuitions that I wanted to explore to their fullest extent. To test them rigorously, I asked Gemini to look for flaws in my reasoning — and it found some good ones.

7.1 — Gemini's four challenges

"To test this theory seriously, I asked Gemini to look for flaws — and he found some good ones:"

1. The Lorentzian signature *Dij is always positive. But relativistic spacetime distinguishes space from time via a specific signature. How does the model distinguish between 'close in space' and 'causally linked'?*

2. The monogamy of entanglement *A system cannot be maximally correlated with an infinite number of others. How can one generate a vast, uniform universe without the fabric collapsing into isolated islands?*

3. The reach of gravity *Gravity acts between two galaxies separated by a near-total vacuum — where Iij is practically zero. How do two systems "know" that they must attract each other without direct correlation?*

4. The $1/r^2$ law *For the theory to be correct, Newton's law must be re-derived from Wij and Deff's dynamics. Does the exponential function used for distance allow for this?*

Gemini's verdict: "Your theory isn't wrong, it's underdetermined. It explains the 'why' beautifully — it still needs to prove the 'how much'."

7.2 — My partial answers

"I haven't answered everything. Newton remains open — I wouldn't know how to do the maths. But here's what I have:"

On the signature (3+1 vs 4) *Our space-time is not 4-dimensional — it is 3+1. The "+1" is not an extra dimension; it is the refresh rate of the correlations. If interactions stopped, space might remain — but time would evaporate instantly. P1 alone in a vacuum: no time, no space. Time is the name we give to the succession of states of interference.*

On granularity *Alice and her atomic clock: 9,192,631,770 oscillations per second, whether the universe is empty or not. Bob moves away at 1/10c — they age differently relative to, but each counts the same number of oscillations. What varies is the relative size of the Planck length itself, one relative to the other. If granularity exists, it is even smaller than Planck.*

On the photon *The photon traverses a universe 'empty of visible matter' — but not empty of structure. It travels through an average of dimensions, a 'renderable' space arising from the decoherence of 10^80 particles. And before arriving, it could have been anywhere — Schrödinger allows for this. The speed c is the speed of propagation within the stabilised 3D grid. Not an absolute limit on information.*

Ehrenfest (1917) demonstrated ***the stability of 3D****: 3 dimensions constitute the stable attractor. But in this model, our 3D is not alone — 1D, 2D, 4D and ND universes coexist. The others are not visible because they are perpendicular to our framework. Not folded — perpendicular.*

On Newton — complete honesty: *"I have not met this challenge. Deriving the $1/r^2$ law again from the dynamics of Wij and Deff requires mathematical tools that I do not master. This is the model's main limitation — and I accept that."*

7.3 — Where it gets dizzying

Black holes as buffer saturation

A black hole, in this context, is a region where the information density becomes so colossal that the MDS embedding can no longer project into 3D. The stress becomes infinite. The 3D "crashes". It is not that space is warped — it is that it can no longer be displayed.

At the event horizon, all the information shifts from the visible sector to the private sector. To the outside observer, the object disappears. In reality, the information isn't lost *(hello Hawking!)* — it's stored in N dimensions that are no longer averaged with our own.

"A black hole is an archive of raw data, uncompressible in 3D."

The singularity is not a point of infinite density — it is a point where Deff explodes. A hole in the sheet, because the sheet is limited to 3D and the singularity is too voluminous in terms of information to fit within it.

Teleportation as metric folding

Speed c is a limit within the stabilised 3D framework. But if two systems increase their entanglement via their private dimensions, their distance Dij tends towards zero — regardless of their apparent separation in the macroscopic rendering.

"Teleportation would not be a movement at infinite speed. It would be a reorganisation of the metric through informational synchronisation. We do not teleport an object through space — we fold space itself."

Einstein hacked

In the classical view, space is a sheet and the particle is a ball placed upon it. In this model, the particle is a knot in the fabric. If one pulls on the thread — velocity, energy, mass — the whole fabric tightens.

"General relativity describes the shape of the smoke. Mutual information theory describes the dynamics of the fire."

7.4 — The spark in the void

The vacuum is everywhere. Infinitely small, infinitely large. It could be in a RAM stick — a hole in the fabric, with enormous potential.

In the vacuum, particles appear and disappear. If two appear simultaneously — local decoherence, a self-contained space-time. Over an infinite timescale, it happens that 10, 100, 10^80 particles synchronise. A universe comes to life. A pixel on a black screen. To us, it is gigantic. On the scale of the primordial vacuum, it is just one more fluctuation.

Our universe is not a creation. It is a resonance that lasted a little longer than the others.

And the primordial void — the chicken and the egg — poses the question we cannot avoid:

“To whom does God address his prayers?”

I don’t have the answer. I would say… an infinite void. One that remains unchanged from what exists today. A spark.

🌟 SECTION 8 — The limits of the model

8.1 Conceptual Limits

Any honest theory knows its limits. Here are mine — acknowledged, documented, and in some cases, disguised as avenues for further research...

1. The exact nature of private degrees

Private degrees are the heart of the model — and, paradoxically, what we are least able to define. Are they real dimensions? Pure information? Environmental modes? The honest answer: we don't know yet. And that's normal at this stage — naming something is not the same as understanding it.

In this model, private degrees play a central role:

- they store information,
- they modulate geometry,
- they may explain dark matter and dark energy.

But their **physical status** remains to be clarified:

- are they real internal dimensions
- informational ancillas
- environmental modes
- local holographic degrees
- hidden quantum variables (in the broad sense)

The model does not decide — and that is to be expected at this stage.

2. Origin of correlated decoherence

In the toy model, γ is a parameter that is set manually. In the real world, nobody turns the knob. What mechanisms produce this decoherence? On what scale? With what intensity? These are open questions — and probably the most important ones for what follows.

The model assumes that decoherence can act:

- on the visible,
- on private individuals,
- or both.

But in the real world:

- what physical mechanisms produce this decoherence
- at what scales
- with what intensities
- in a correlated or uncorrelated manner

These questions remain open.

3. Absence of an explicit metric

The model reconstructs a geometry **from informational distances**, but it does not yet derive:

- a continuous metric,
- a connection,
- a curvature,
- field equations.

This is the main technical limitation — and the most promising avenue of research.

4. No complete geometric dynamics yet

We observe:

- emergence,
- deformation,
- collapse.

But we do not yet have:

- an equation describing the evolution of the geometry,
- an analogue of Einstein's equations,
- a conservation of geometric information.

8.2 Numerical limitations

The limitations are not merely conceptual — they are also practical. The toy model can be simulated, but it is resource-intensive, and it is small.

1. System size

The toy model uses 4 visible particles. This is sufficient to illustrate the phenomena, but insufficient for:

- complex structures,
- extensive networks,
- realistic cosmological phenomena.

2. Exponential cost

Each particle has several private ancillas. The total size grows exponentially:

dim(H)=2nvis+npriv

In plain terms: doubling the number of particles does not simply double the computation time — it multiplies it exponentially. This is why the tensor networks (MPS, MERA) mentioned in the roadmap are a serious avenue to explore: they circumvent this problem

3. Embedding limited to 2D

MDS reconstructs a 2D or 3D geometry, but:

- the actual geometry could be more complex,
- certain structures are not isometric in low dimensions,
- stress can mask subtle phenomena.

2D is what we can easily visualise — but our universe is 3+1. 3D embedding is possible, but 4D is much less readable. This is a limitation of both display and computation.

8.3 Interpretative limitations

The cosmological interpretations in Section 6 are consistent with the model — but consistent does not mean proven. This is speculative physics, not established cosmology.

1. Speculative cosmology

The interpretations:

- informational Big Bang,
- dark matter = private correlations,
- dark energy = loss of visible correlations,
- gravity = information gradient,

are consistent, but remain **speculative**. They require:

- larger models,
- quantitative comparisons,
- links with observations.

2. Ambiguity between correlation and causality

The model states:

> “correlation = proximity”

But it does not yet state:

> “proximity = causality”

A finer temporal structure is missing.

This is perhaps the most profound conceptual limitation: two correlated particles are 'close' in this model. But close does not mean causally linked. A finer temporal structure is needed to distinguish between 'they resemble each other' and 'one has influenced the other'.

8.4 Roadmap: realistic avenues of research

What follows is not a programme I intend to carry out alone — I have neither the tools nor the training. It is a roadmap for someone who might wish to continue. If this document falls into the right hands, here is where to start.

Step 1 — Extend the toy model

- increase the number of visible ones from 4 to 6 or 8,
- increase the number of ancillas,
- test different initial states,
- explore other types of decoherence (amplitude damping, depolarisation).

Step 2 — Study informational percolation

- precisely identify the threshold γc,
- to study the nature of the transition,
- compare with classical phase transitions.

Step 3 — Construct an effective metric

Starting from Dij, attempt to derive:

- a metric $g\mu\nu$,
- a connection,
- a curvature,
- a geometric dynamica.

Step 4 — Link the model to emergent gravity

Compare with:

- Verlinde (entropic gravity),
- Jacobson (Einstein = thermodynamics),
- Van Raamsdonk (entanglement ↔ connectivity).

Step 5 — Explore the cosmological implications

- model a simple "informational universe",
- simulate expansion via the loss of correlations,

- test the hypothesis “dark matter = private correlations”.

Step 6 — Move to tensor networks

The model could be reformulated in terms of:

- MPS (Matrix Product States),
- MERA (Multiscale Entanglement Renormalisation Ansatz),
- PEPS (Projected Entangled Pair States).

These tools are **inherently geometric**.

Step 7 — Formalising a geometric dynamics

Ultimate goal:

> to derive an equation of evolution for the emerging geometry from the dynamics of correlations.

Step 7 is the ultimate goal — an equation of evolution for the emerging geometry. It is ambitious. It may be 10 years of serious research away. But the path has been laid out.

🌟 SECTION 9 — Appendices

Appendix A — Detailed numerical protocol

This appendix describes how to reproduce the toy model simulations.

A.1 Construction of the initial state

1. Create a local GHZ for each particle: $|000\rangle+|111\rangle$ on (vis, a, b).
2. Create two Bell pairs between visible particles: (0–1) and (2–3).
3. Add a slight phase noise to break the symmetries.

A.2 Decoherence

Use Lindblad’s master equation:

$$d\rho dt=-i[H,\rho]+\sum k(Lk\rho Lk\dagger-12\,Lk\dagger Lk,\rho)$$

Where:

- H=0 or single coupling,
- Lk=γkZk.

A.3 Diagnostics

At each time step:

- reduce the states,
- calculate the entropies,
- calculate Iij,
- normalise to W,
- calculate Deff,
- construct Dij,
- MDS embedding,
- stress,
- half-life.

A.4 Exporting data

Export:

- Iij(t) matrices,
- Dij(t) matrices,
- W spectra,
- embeddings,
- Deff(t) curves,
- t1/2(γ) curves.

Appendix B — Enhanced debugging checklist

✔ Check the symmetry of I

$Iij = Iji$

✔ Check the diagonal

$Iii = 0$

✔ Check the normalisation of W

Each row must sum to 1.

✔ Check the eigenvalues

They must be real (or nearly so).

✔ Check the stress

Stress too high → unreliable embedding.

✔ Check the limits

$\gamma \to 0$ → stable geometry $\gamma \to \infty$ → collapse

🌟 SECTION 10 — References

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